

WASP-140b

R-2134

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_230131 · created 2026-08-11T23:43:38+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459975.6512 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.017
Transit depth (Rp/Rs)^2	1.87 +/- 0.47 [%]
Orbital Inclination (inc)	83.89 +/- 0.44
Airmass coefficient 1 (a1)	1.2672 +/- 0.0089
Airmass coefficient 2 (a2)	-0.124 +/- 0.0053
Scatter in the residuals of the lightcurve fit is	0.7 %
Best Comparison Star	None
Optimal Aperture	4.09
Optimal Annulus	8.48
Transit Duration (day)	0.0682 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.137 ± 0.017 vs 0.1432 (deviation 0.36σ)
Duration fit vs published	0.0682 d vs 0.0604 d (deviation 1.21σ)
Mid-time O–C	0.9 min
Depth SNR	8.0
Residual scatter	0.7 %

Light curve

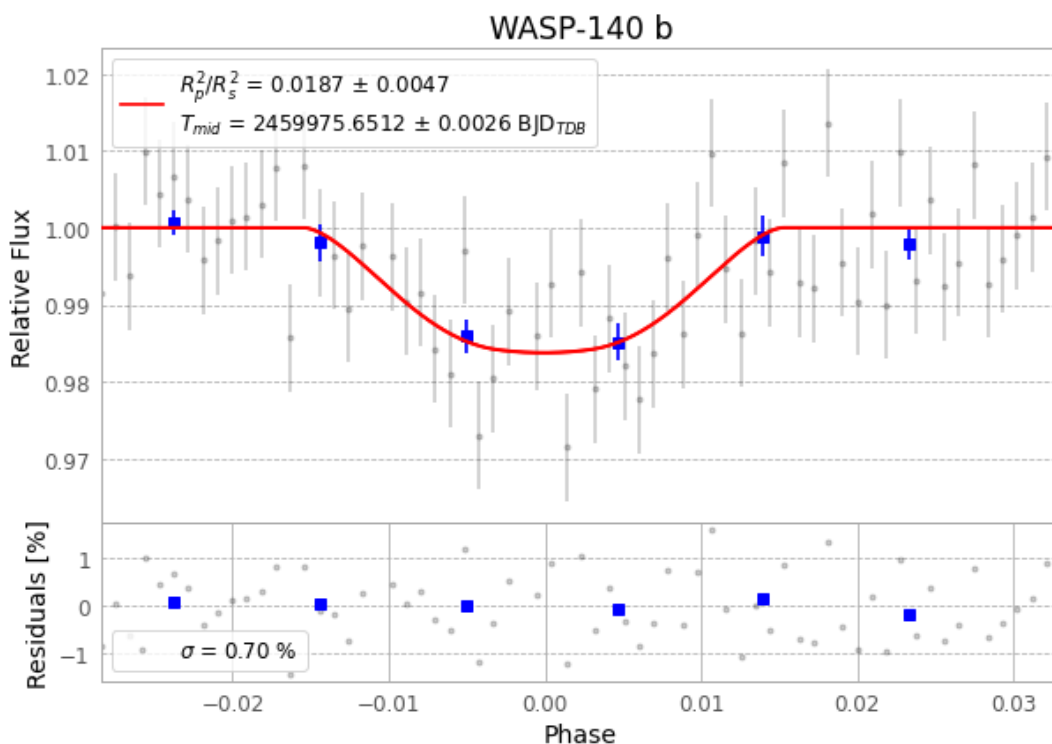


Figure 1 — FinalLightCurve_WASP-140 b_2023-01-30.png (SHA-256 33e773da47e75d96...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 171], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1e8ad46523c546fe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

67 FITS frames (44 MB), WASP-140230131020315.FITS → WASP-140230131052115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-230131.log (03258141eb7a...), FinalLightCurve_WASP-140 b_2023-01-30.png (33e773da47e7...), FinalLightCurve_WASP-140 b_2023-01-30.csv (705619642fb3...)

Compute resources

Wall time 172.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:43Z.